PHONEPE TRANSACTION INSIGHTS DASHBOARD

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1. Title Page

Project Title:
PhonePe Transaction Insights Dashboard

Submitted By:
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Technology:
Python | MySQL | Streamlit | Data Visualization | Data Analysis | ETL (Extract, Transform, Load) | Plotly

Date:
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2. Introduction

This project focuses on analyzing PhonePe transaction data to gain insights into user behavior, transaction trends, and geographical patterns. The dataset is processed using Python and stored in MySQL. Interactive visualizations are created using Streamlit and Plotly to build a dynamic dashboard.

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3. Objectives

- To analyze digital transaction patterns across India
- To identify top-performing states and categories
- To visualize data using charts and maps
- To build an interactive dashboard using Streamlit
- To derive business insights from transaction data

Tools & Technologies

- Python
- Pandas
- MySQL
- Streamlit
- Plotly
- ETL (Extract, Transform, Load)
- Data Visualization
- Data Analysis

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5. Data Description

The dataset is taken from PhonePe Pulse and consists of:

🔸 Aggregated Tables

- aggregated_transaction
- aggregated_user
- aggregated_insurance

🔸 Map Tables

- map_transaction
- map_user
- map_insurance

🔸 Key Fields

- state
- year
- quarter
- transaction_type
- transaction_amount
- transaction_count

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6. Data Processing

- JSON files were read using Python

- Folder structure was traversed using `os.listdir()`
- Data was cleaned and structured
- Missing values were handled
- Data was inserted into MySQL tables
- SQL queries were used for aggregation

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7. Data Analysis

The following analysis was performed:

- Top 10 states by transaction amount
- Transaction category distribution
- District-wise transaction trends
- Insurance growth analysis
- User engagement through app opens

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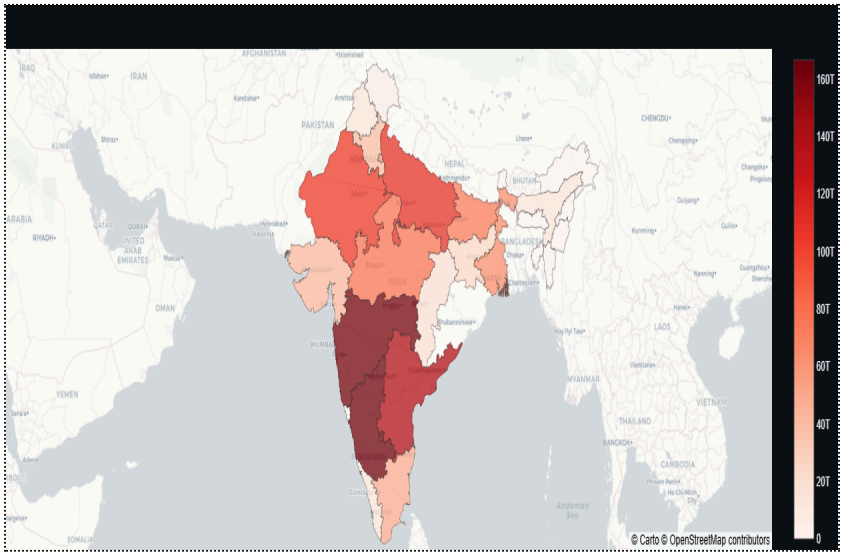
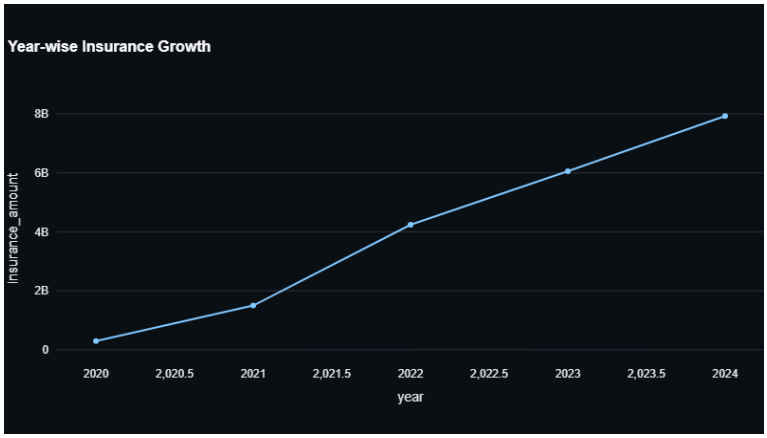
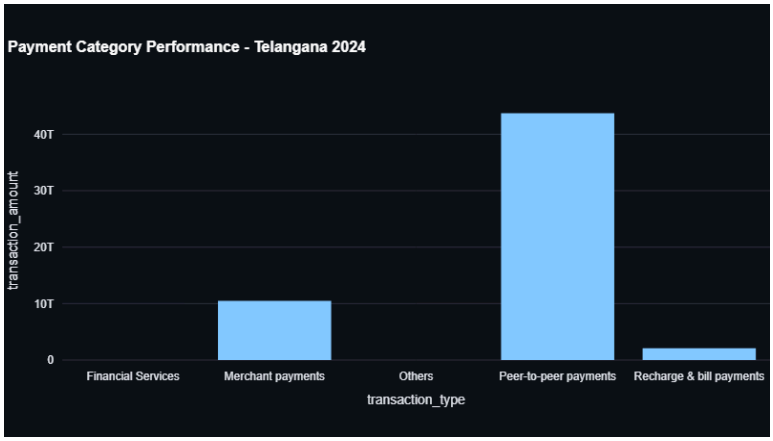
8. Visualizations

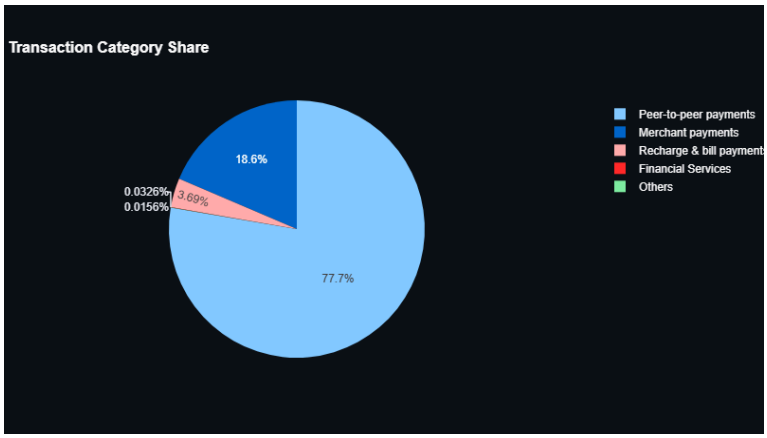
The project includes the following visualizations:

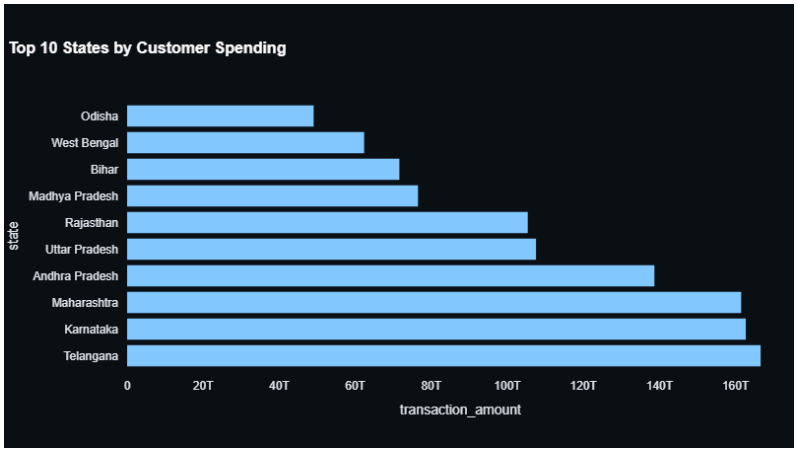
1. Bar charts (Top states, districts)
2. Pie charts (Transaction categories)
3. Line charts (Year-wise trends)
4. Choropleth map (Geographical analysis)




Year-wise Insurance Growth
8B
6B
4B
2B
0
insurance_amount
2020
2,020.5
2021
2,021.5
2022
2,022.5
2023
2,023.5
2024
year


Payment Category Performance - Telangana 2024
40T
30T
20T
10T
0
transaction_amount
Financial Services
Merchant payments
Others
Peer-to-peer payments
Recharge & bill payments
transaction_type


Transaction Category Share
18.6%
0.0326%
0.0156%
3.69%
77.7%
Peer-to-peer payments
Merchant payments
Recharge & bill payments
Financial Services
Others



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9. Business Use Cases (IMPORTANT)

- **Customer Segmentation**
- **Payment Performance**
- **Geographical Insights**
- **User Engagement**
- **Insurance Insights**

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1 Customer Segmentation

Customers are segmented based on transaction activity across states.
Telangana has the most transactions compared with other states, High transaction states indicate strong user engagement and can be targeted for premium services.

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2 Payment Performance

Different transaction categories (Recharge, Bills, etc.) are analyzed to understand user preferences and popular services.

3️⃣ Geographical Insights

The choropleth map shows transaction distribution across states.
States with higher transaction amounts indicate stronger adoption of digital payments.

4️⃣ User Engagement

User activity is measured using app opens and transaction frequency.
Higher engagement indicates better user retention.

5️⃣ Insurance Insights

Insurance Insights focuses on analyzing insurance-related transaction data from PhonePe. It helps identify states and districts with higher insurance adoption and shows how insurance transactions grow over time. This insight can help businesses improve insurance product offerings and target regions where insurance awareness is low.

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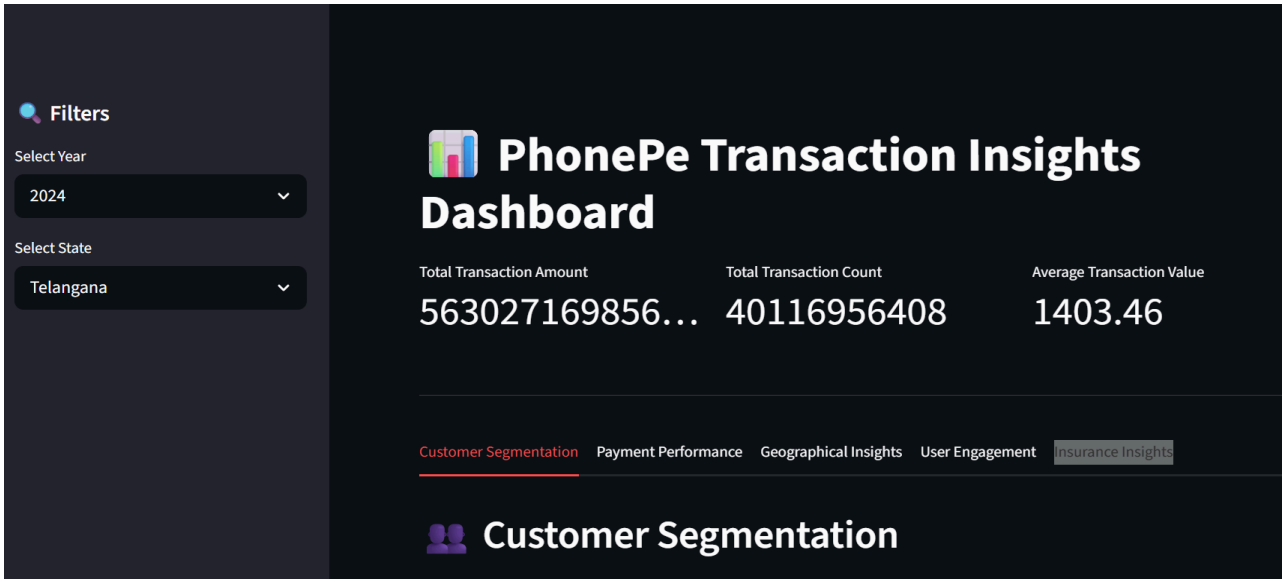
10. Dashboard (Streamlit)

An interactive dashboard was built using Streamlit with:

- Sidebar filters (Year, State)
- KPI metrics (Amount, Count, Average)
- Multiple tabs for analysis
- Interactive charts and maps



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11. Map Interpretation

The choropleth map represents state-wise transaction data.

- Darker color → Higher transaction amount
- Lighter color → Lower transaction amount

This helps identify regions with strong digital adoption and areas needing improvement.

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12. Conclusion

The project successfully analyzes PhonePe transaction data and provides meaningful insights into user behavior, geographical trends, and transaction patterns. The dashboard makes it easy to explore and understand large datasets interactively.